RNS INSTITUTE OF TECHNOLOGY, BENGALURU — DEPARTMENT OF ECE

Device-Free Human Activity Recognition & Fall Detection Using WiFi Channel State Information

COTS ESP32-S3 Hardware • 200 Hz HT40 Link • Rigorous LOSO Cross-Validation • Real-Time Engine

PROJECT GROUP 10 (2026-2027)

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Motivation & Problem Statement: The Case for WiFi CSI

Speaker 1

1. Camera-Based Vision Systems

Major Drawbacks:

- Severe privacy invasion in private domestic spaces (bathrooms, bedrooms).
- Strict line-of-sight requirement; completely blocked by walls and smoke.
- Fails in low-light, darkness, or intense optical glare.
- High compute bandwidth required for continuous high-res video streaming.

2. Wearable Sensor Nodes

Major Drawbacks:

- Requires active user compliance and routine battery replenishment.
- Elderly adults and dementia patients frequently forget or remove devices.
- Removed during sleep or bathing (where over 60% of senior falls occur!).
- Skin irritation, discomfort, and risk of loss/damage.

3. Proposed WiFi-CSI Sensing

Core Engineering Advantages:

- 100% Privacy-Preserving: Captures zero optical or acoustic images.
- Through-Obstacle Penetration: 2.4 GHz RF penetrates non-metallic partitions.
- Device-Free: Operates completely contactlessly; user wears zero hardware.
- Ultra-Low Cost: Built on commodity ESP32 hardware (~$15 total).
- Dual-Use Infrastructure: Leverages ubiquitous ambient wireless fields.

RF Physics & CSI Fundamentals: Physical Principle & Multipath

Speaker 1

OFDM Channel Estimation Model

Frequency-Domain Channel Response

- In 802.11n OFDM, the received symbol Y(f,t) relates to transmitted symbol X(f,t) via:
- $Y(f, t) = H(f, t) \cdot X(f, t) + N(f, t) \implies H(f, t) = Y(f, t) / X(f, t)$
- H(f,t) is the complex Channel Frequency Response (CFR), estimated across subcarriers from known Long Training Field (LTF) symbols in every packet preamble.

Multipath Superposition & Motion

Constructive / Destructive Interference

- $H(f,t) = H_{static}(f) + H_{dynamic}(f, t)$ = sum of L physical propagation paths.
- Static paths reflect off walls, floor, and metal cabinets; dynamic path reflects off moving limbs.
- Doppler Shift: $f_D = (1 / \lambda) \cdot (d/dt)[d_{dynamic}(t)]$. At 2.462 GHz, $\lambda \approx 12.2$ cm.
- Sub-Centimeter Sensitivity: Moving a body by $\lambda/4 \approx 3$ cm flips the reflected phase by 180°.
- CSI vs RSSI: RSSI is a single coarse scalar power value; CSI provides fine-grained complex measurements across 114 subcarriers simultaneously.

Experimental Hardware Architecture & Library Testbed Layout

Speaker 1

Hardware Component Architecture

Off-The-Shelf Transceivers

• Transmitter (TX): NodeMCU-32S (ESP-WROOM-32U, Xtensa LX6 @ 240 MHz).
• TX Antenna: External 8 dBi directional patch antenna mounted at height h = 0.9 m at (0.0, 0.0) m.
• Receiver 1 (RX1): YD-ESP32-S3 (Xtensa LX7 @ 240 MHz) + 6 dBi omni antenna at h = 1.0 m at (-1.3, 4.5) m.
• Receiver 2 (RX2): YD-ESP32-S3 + 6 dBi omni antenna at h = 0.5 m at (+1.3, 4.5) m.
• Host Node: Linux laptop (Intel i3-7020U, 7 GB RAM) receiving serial streams at 2,000,000 baud.

Room Geometry: College Library Hall

Coordinate Frame (position_map.yaml v2)

• Room: College Library Main Hall (active sensing area: 4.5 m x 5.0 m).
• Chair A: (-0.9, 3.2) m — sits directly on the direct RX1 line-of-sight link.
• Chair B: (+1.0, 1.5) m — sits directly on the direct RX2 line-of-sight link.
• Chair C: (0.0, 1.2) m — central chair positioned near the transmitter.
• Mattress M: Bounded area x in [-0.5, 1.5], y in [2.9, 3.8] m (all simulated falls performed here).
• Path 1 (Vertical): Walk along the LoS link (x = -1.5, y in [1.2, 4.0]).
• Path 2 (Horizontal): Walk across the LoS link (y = 2.55, x in [-2.0, 2.0]).

Radio Layer Specifications & Binary Wire Framing Protocol

Speaker 1

Radio Layer Physical Parameters

IEEE 802.11n Configuration

• Bandwidth / Channel: Wi-Fi Channel 11 (2462 MHz), HT40 mode (40 MHz width, 2432–2472 MHz).
• PHY Rate: Fixed to MCS0 (BPSK, 1/2 code rate) with Long Guard Interval (800 ns). Zero rate adaptation.
• Packet Transmission Rate: Exactly 200 Hz (5000 µs hardware esp_timer interrupt period).
• TX Output Power: Configured at 5 dBm (CONFIG_TX_POWER_QDBM = 20), saved in NVS (D-017, D-029).
• Transmission Mode: Connectionless ESP-NOW broadcast with static radio_id = 0xCBD36978.

Wire Protocol v2 Framing

High-Throughput Serial Streaming

• Frame Format: 48-byte packed C-struct header framed by sync bytes 0xA5 0x5A.
• Data Integrity: Protected by CRC-16/CCITT-FALSE across header and payload bytes.
• Active Carriers: 384 raw CSI bytes per packet representing HT-LTF I/Q values.
• Audit Verification: Decoded 250 files across 125 morning takes with ZERO CRC errors (D-011 limit < 0.5%).
• Board Liveness: Firmware v2.1 introduces task watchdog and boot reset reasons in HELLO frames.

Data Collection Protocol, Safety Procedures & RF Hygiene

Speaker 2

Experimental Recording Protocol

Controlled Script Execution

• Acoustic Pacing: Automated terminal speech countdown ensured strict protocol synchronization.
• Event Takes (10 s): 3 s motionless baseline -> Spoken 'GO' prompt -> Action execution (1–2 s) -> 4 s settling posture.
• State Takes (30 s): Continuous, steady-state execution (sitting, standing, walking, lying, or empty).
• Fall Safety Mandate: All falls performed onto a high-density gymnasium mattress with spotters present.
• Action Look-Alikes: Included sit_down, stand_up, and bend_pick to challenge the fall detector.

RF Hygiene Protocol (Decision D-032)

Eliminating In-Band Contention

• The Contention Discovery: Link check before S5 revealed that leaving host laptop Wi-Fi active caused 41.5% packet loss on RX2 at strong RSSI (-38 dBm) due to local radio contention.
• Hygiene Rule Enforced: Host laptop onboard Wi-Fi and Bluetooth disabled via rfkill; all bystander smartphones switched to airplane mode.
• Impact: Disabling laptop radios instantly dropped packet loss to 1.1% on RX1 and 4.7% on RX2.
• Rule Codified: Made mandatory for all subsequent recording and live system operations.

The Curated Dataset & Phase 0 Audit: 202 Takes Across 6 Folds

Speaker 2

Final Dataset Composition

202 Recorded Takes (801 Files Total)

- Cohort: 6 healthy adult volunteers (S1 to S6) + unoccupied room takes (none).
- Morning Session (S1–S4): 125 takes (118 subject takes + 4 baseline empty takes).
- Afternoon Session (S5–S6): 77 takes recorded after an intentional transmitter realignment.
- Activity Breakdown: 42 falls, 43 sit-downs, 46 stand-ups, 20 bend-and-picks, 16 walks, 11 stands, 11 sits, 6 lies, 7 empty.
- Format 2 Windows: Exactly 2401 windows of 4 s extracted (hop 0.5 s, 20% quality gate).

Session Effect & Audit Findings

Phase 0 Read-Only Audit (docs/AUDIT.md)

- Audit Immutability: Scripts read data/raw as read-only; 801 files verified by SHA-256 manifest.
- Take Overrides (D-018): S1_T29_walk was marked invalid due to operator abort; decoded data was intact and overridden via config/take_overrides.csv without editing raw meta.json.
- Session Drift (D-033): S5/S6 show no RSSI drift (≤ 1.0 dB shift), but static channel shape shifted 3.4–4.4 dB RMS between morning and afternoon sessions due to TX re-aim.
- Evaluation Impact: S5 and S6 represent true cross-session test folds.

DSP Pipeline: Subcarrier Layout, Sequence Alignment & Gap-Fill

Speaker 2

Active HT-LTF Subcarriers

114 Used Subcarriers Verified

• 802.11n 40 MHz Channel Layout: 128 total subcarrier bins in FFT.
• Active Energy Carriers: Verified empirically across 396 receiver takes:
• k in [-58, -2] and [+2, +58] ==> Exactly 114 active subcarriers.
• Null Subcarriers: DC carrier (bin 0, ±1) and guard bands ([-64..-59] and [+59..+63]) carry zero signal.
• L-LTF Bins: Evaluated in Phase 2; confirmed to occupy upper 20 MHz only. Dropped to focus on 114 HT-LTF.

Packet Synchronization & Linear Gap-Fill

Phasor-Domain Interpolation

• Sequence Alignment: RX1 and RX2 streams are aligned by the TX sequence number embedded in firmware.
• Linear Gap-Fill: Missing packets within acceptable burst limits are reconstructed via linear interpolation across the complex phasor domain.
• Quality Gate (D-039): A window is rejected if either receiver suffers > 20% missing packets or any single gap exceeds 100 ms (20 packets).
• Diagnostic Impact: Relaxing initial strict 10% gate to 20% reduced window drops from 27.8% down to 3.9%, recovering 67% of S1's otherwise lost data without distorting DSP.

DSP Pipeline: FIR Anti-Aliasing, Decimation & Gain Steps

Speaker 2

Anti-Aliasing FIR Decimation

Suppressing High-Frequency Out-Of-Band Noise

• Human Kinetic Limits: Normal human body motion generates Doppler shifts bounded below 30–40 Hz (v ≈ 2 m/s, f_D ≈ 32.8 Hz).
• Anti-Alias FIR Filter: 27-tap linear-phase Equiripple low-pass filter (passband 0–35 Hz, -45 dB stopband attenuation at 50 Hz).
• Decimation: Downsampled from raw 200 Hz to 100 Hz, eliminating high-frequency serial noise and spectral folding.
• Phase Coherence: Linear-phase FIR design introduces constant group delay without phase distortion.

Resolving 6.1 dB Automatic CSI Gain Steps

Physics of ESP-IDF AGC Steps (D-035)

• The Symptom: CSI amplitude exhibited alternating 6.1 dB power drops uncorrelated with RSSI, heavily concentrated at Chair B.
• Root Cause: ESP32-S3 internal automatic CSI scaling (manu_scale = false) halving int8 digital resolution under high signal levels.
• The Fix: Per-packet gain normalization divides each subcarrier by the packet's mean amplitude:
• X_norm(k, t) = X_raw(k, t) / [(1/114) * sum_i(X_raw(i, t))]
• Integrity Rule: Low-scale packet fraction is permanently banned as a feature (correlates with Chair B position).

DSP Pipeline: Subcarrier Phase Unwrapping & Sanitization

Speaker 2

The Raw Phase Problem

Carrier Frequency & Packet Delay Noise

• Raw Phase Instability: Raw phase angles exhibit uniform random distribution across packets (circular std ≈ 2.82 rad).
• Sources of Corruption: Caused by Carrier Frequency Offset (CFO), Sampling Frequency Offset (SFO), and microsecond Packet Detection Delays (PDD).
• Equation of Corruption: $\theta_{raw}(k, t) = \theta_{true}(k, t) - 2\pi \cdot k \cdot \Delta t / N + \beta + Z_{noise}$.

Two-Stage Linear Sanitization (Decision D-036)

Subcarrier Unwrapping & Detrending

• Stage 1: Frequency Unwrapping across subcarriers: $\theta_{unwrap}(k, t) = \text{unwrap}(\theta_{raw}(k, t))$.
• Stage 2: Least-Squares Linear Detrending to remove slope α_t and intercept β_t for each individual packet:
• $\theta_{sanitized}(k, t) = \theta_{unwrap}(k, t) - (\alpha_t \cdot k + \beta_t)$
• Quantitative Noise Reduction: Sanitized phase circular standard deviation collapses from 2.82 rad down to 0.13–0.16 rad!
• Physical Utility: Unveils path-length variations; dynamic phase features weighted by amplitude squared to neutralize 2π slips.

EXPERIMENTAL RIGOR

The Spatial Leakage Trap: Discovering the Position Confound

Speaker 2

The Hidden Experimental Trap

Why Standard ML Pipelines Cheat

• Spatial Confounding (D-024): In home monitoring, actions naturally correlate with furniture (falls on mattress, sitting at chairs, walking on paths).
• Body Shadowing Effect: Sitting at Chair A shadows RX1; sitting at Chair B shadows RX2.
• The Leakage: Static features (mean RSSI, static subcarrier band amplitudes) encode WHERE a person is rather than WHAT action is occurring.
• The Academic Fallacy: Published papers using static features achieve 95% accuracy because the model simply memorizes room coordinates.

Our Architectural Solution: Static vs. Dynamic Split

Enforcing Genuine Kinematic Sensing

• Bifurcated Feature Architecture: Every feature explicitly classified into Static (Spatial) or Dynamic (Kinematic) groups.
• Mandatory Dynamic-Only Reporting: All primary event models are trained strictly on dynamic motion features (temporal variance, Doppler powers).
• Empirical Proof of Confound: Static features alone drop event macro-F1 to 0.353; identical sit-down actions at Chair A achieved 76% recall, but collapsed to 25% at Chair B where 16–22% were mistaken for falls!
• Rigorous Reporting: Position confounding documented as a first-class project limitation in REPORT.md.

Feature Engineering: The 83-Dimensional f2 Feature Space

Speaker 2

Static Spatial Group (33 Features)

Captures Static Multipath Profile

• Band Mean Amplitudes (16): Normalized mean amplitude across 8 subcarrier sub-bands for RX1 and RX2.
• Received Power Levels (3): Mean RSSI for RX1, RX2, and cross-receiver spatial difference (RX1 - RX2).
• Spatial PCA Loadings (14): Explained variance ratios of top 3 principal components and carrier quarter loadings.
• Utility: Used for baseline presence detection; excluded from dynamic event classification.

Dynamic Motion Group (50 Features — Primary Input)

Captures Doppler Kinematics

• Temporal Fluctuations (12): Standard deviation and 10 ms absolute differences across subcarriers (RX1/RX2).
• Sliding Motion Energy (8): 0.25 s sliding standard deviation of PC1–3 and temporal peak location.
• Doppler Spectral Bandpowers (12): FFT bandpowers of PC1: sway (< 1 Hz), walking (1–3 Hz), and fall impact (> 3 Hz).
• Inter-Carrier Coherence (2): Mean correlation across subcarrier fluctuations; high coherence indicates macroscopic body movement.
• Sanitized Phase Dynamics (10): Amplitude-weighted circular variance and phase derivative energy.
• Cross-Receiver Dynamics (4): Log-ratios of temporal variance and cross-correlation of RX1/RX2 motion envelopes.

Evaluation Methodology: Leave-One-Subject-Out (LOSO)

Speaker 3

Why Random Cross-Validation Fails

Eliminating Subject Data Leakage

• The Random Split Flaw: Shuffling windows randomly into 80/20 train/test leaks identical subject multipath and clothes reflections into both splits.
• The LOSO Standard: Evaluation is conducted strictly via 6-Fold Leave-One-Subject-Out (LOSO) cross-validation across all subjects S1 to S6.
• Complete Isolation: In each fold, all takes from one volunteer are completely withheld from training and used strictly as test data.
• Round-Robin Baselines: The 7 empty-room takes are distributed round-robin across folds (D-028) so every fold maintains true negative presence examples.

Nested Hyperparameter Tuning

Zero Test Fold Contamination

• Nested Search: Within each outer training fold, hyperparameter optimization uses GroupKFold(n_splits=3) grouped by training subjects.
• Feature Scaling: StandardScaler fitted strictly on training fold distributions; test fold transformed without leakage.
• Scoring Metric: Macro-averaged F1-score across folds (Mean ± Std).
• Reporting Standard: Deep models evaluated across 3 random seeds per fold to guarantee statistical repeatability.

Temporal Window Length Optimization: 2 s vs 3 s vs 4 s

Speaker 3

Objective Optimization Rule

Pre-Registered Selection Metric (D-046)

• Window Selection Rule: Average LOSO macro-F1 across presence, state, and event tasks using dynamic features.
• Evaluated Lengths: 2.0 s, 3.0 s, and 4.0 s sliding windows with 0.5 s step hop.
• 2-second Windows: Score = 0.533 (Events: 0.450, State: 0.563, Presence: 0.585).
• 3-second Windows: Score = 0.582 (Events: 0.503, State: 0.580, Presence: 0.665).
• 4-second Windows: Score = 0.603 (Events: 0.539, State: 0.572, Presence: 0.698)
==> WINNER (4 s Adopted).

Physical Kinematic Justification

Capturing the Anatomy of a Fall

• Anatomy of a Fall: A human fall consists of: (1) Pre-fall loss of balance, (2) Rapid downward descent (0.5–1.0 s), (3) Floor impact shock, and (4) Post-fall motionless immobility.
• Window Sizing Physics: A 2 s window truncates either the initial descent or post-fall stillness.
• The 4 s Advantage: A 4 s window captures the entire transition from standing motion to impact to lying prone.
• Latency Trade-off: While 4 s introduces slightly longer window latency, a 0.5 s hop rate ensures the live engine refreshes predictions every 500 ms.

MODEL COMPARISON

Model Benchmarking (Track A): Classical Machine Learning

Speaker 3

Benchmark Results on 4 s Windows (Dynamic f2 Features)

LOSO 6-Fold Performance Summary

• Events (5 Classes: Fall, Sit, Stand, Bend, None):
• • SVM-RBF (Dynamic f2): Macro-F1 = 0.539 ± 0.054 (Fall F1: 0.521, Fall Recall: 0.552).
• • XGBoost (Dynamic f2): Macro-F1 = 0.523 ± 0.084 (Fall F1: 0.560, Fall Recall: 0.512).
• • LightGBM / Extra Trees: Macro-F1 = 0.505 / 0.502.
• • Random Forest Baseline: Macro-F1 = 0.486 ± 0.071.
• Postural State (4 Classes: Sit, Stand, Walk, Lie):
• • Extra Trees (300 trees): Macro-F1 = 0.572 ± 0.115 (Best performing model).
• Presence Detection (2 Classes: Empty vs Occupied):
• • Logistic Regression: Macro-F1 = 0.698 ± 0.065 (Sensitivity: 0.855, Specificity: 0.828).

Analytical Observations

Key Insights

• Linear vs Kernel Boundaries: SVM with Radial Basis Function kernel excelled at complex dynamic event boundaries.
• Ensemble Robustness: Extra Trees outperformed single trees and standard RF on stationary postures.
• Cross-Session Stability: S5 and S6 afternoon folds achieved strong F1 (0.595 and 0.520 on SVM events), confirming dynamic features generalize across sessions.

Benchmarking Track B (MiniRocket) vs Track C (Deep Neural Nets)

Speaker 3

Track B: MiniRocket Transform

NumPy Re-Implementation (D-043)

• Architecture: 10,000 random convolutional kernels mapping 16 band channels at 100 Hz to pooled features with a Ridge Regression head.
• Events Macro-F1: 0.572 ± 0.062 (Rank 1 by selection rule D-047).
• Presence Macro-F1: 0.729 ± 0.178 (Highest raw mean, but high variance across folds).
• The MiniRocket Flaw: Poor window-level fall recall (0.434 vs 0.552 for SVM); uncalibrated ridge scores; high CPU inference latency (326 ms single-window).

Track C: 10 Deep Learning Architectures

Fixed Budget (AdamW, Patience 6, 3 Seeds)

• Evaluated Models: 1D-CNN, CNN-LSTM, CNN-BiLSTM, BiLSTM, GRU, TCN, ResNet-1D, ABLSTM, THAT (Transformer). Model sizes: 19k to 127k parameters.
• Events Performance: Best deep model (CNN-BiLSTM) achieved only 0.344 ± 0.041 macro-F1.
• Remaining Architectures: Scored between 0.332 and 0.264 macro-F1.
• Outcome: Deep learning models severely underperformed classical tabular models by over 20% F1.

FAILURE ANALYSIS

Why Deep Learning Failed: Empirical Loss Curves & Small Data

Speaker 3

The Overfitting Divergence

Training vs. Validation Trajectories

- Empirical Evidence: Recorded across all 10 architectures (results/phase4/track_c/*_curves.png).
- Training Loss: Rapidly plunged to ~0.2 within 5 epochs.
- Validation Loss: Diverged sharply upwards beginning as early as epoch 3!
- Early Stopping: Consistently triggered at median epochs 5 to 9 across all folds.
- Event Class Collapse: Deep models severely under-predicted transient events, exhibiting recalls between 0.10 and 0.40.

Physical Root Causes of Deep Failure

Theoretical Lessons

- Small Sample Regime: 2401 windows from 6 human subjects is insufficient to constrain 100,000+ deep neural parameters.
- Multipath Domain Shift: Deep networks memorized the static multipath scattering profiles of individual training subjects rather than generalizable body kinematics.
- Tabular Dominance: Physics-engineered features (Doppler power, phase circular variance) combined with max-margin classifiers (SVM) provide essential inductive bias.
- Scientific Honesty (D-048): Deep failure reported transparently as-run without post-hoc cherry-picking.

DECISION REGISTER D-050

Gate 4 Model Selection: Justifying Departures from the Rule

Speaker 3

Adopted Final Production Models

4 s Windows, Dynamic f2 Features

• Events Task: SVM-RBF (C = 10, γ = 0.01) — Macro-F1: 0.539 ± 0.054.
• Postural State Task: Extra Trees (300 trees, max_feat = 0.3) — Macro-F1: 0.572 ± 0.115.
• Presence Task: Logistic Regression (C = 1.0) — Macro-F1: 0.698 ± 0.065.
• Fall Alarm Engine: Platt-Calibrated Binary SVM-RBF (C = 1.0, γ = 0.01).

Why We Overruled MiniRocket (Decision D-050)

Life-Safety & Real-Time Rationale

• 1. Fall Sensitivity Priority: MiniRocket achieved higher overall F1 (0.572 vs 0.539) by predicting common classes, but had POOR fall recall (0.434 vs 0.552 for SVM). For a fall alarm, missing falls is unacceptable.
• 2. Probability Calibration: MiniRocket's ridge head outputs uncalibrated scores; SVM provides Platt-calibrated posterior probabilities P(fall) essential for threshold gating.
• 3. Real-Time Latency: Single-window MiniRocket transform requires 326 ms on CPU; SVM inference requires < 1 ms on f2 features (total Predictor call = 67 ms with cached PCA), easily fitting inside 500 ms engine cycles.
• 4. Presence Stability: MiniRocket collapsed on subjects S3 (0.474) and S4 (0.487); Logistic Regression was completely stable (± 0.065).

The Calibrated Fall Alarm Chain Architecture (Decision D-051)

Speaker 3

Continuous Stream Decision Chain

Moving Beyond Isolated Window Testing

• The Reality Gap: Testing isolated windows does not reflect reality. In real life, a streaming engine processes continuous overlapping windows.
• Binary Fall Detector: Dedicated Binary SVM-RBF (fall vs rest) with Platt (sigmoid) calibration fitted strictly on inner training folds.
• Nested Threshold Tuning: Threshold θ (0.30–0.95) and sliding window rule (k-of-n) optimized on inner training LOSO streams.
• Adopted Production Rule: Alarm triggered if ≥ 2 of the last 3 windows have P(fall) ≥ 0.50.
• Alarm Cooldown: 10-second lockout timer prevents duplicate burst alarms following an initial trigger.

Live = Offline Equivalence (Decision D-054)

Zero Train/Serve Skew

• Verification Test: Replayed 12 real takes packet-by-packet through the live buffer and compared outputs against the offline pipeline.
• Bit-Identical Result: Exact 0.00000 maximum absolute difference in features, X, P, and probabilities across all 378 real windows!
• Significance: Proves zero train-serve skew between laboratory Python code and production streaming runtime.

Fall Detection Stream Performance: Evaluating the Full Chain

Speaker 3

Continuous Stream Benchmark Results

Tested on All 202 Real Takes (2401 Windows)

- Metric | Baseline 5-Class SVM (θ=0.6, 2-of-3) | Adopted Calibrated Chain (D-051)
- ----------
- Falls Detected | 18 of 42 (42.9%) | 21 of 42 (50.0%)
- Look-Alike False Alarms | 9 of 109 takes (8.3%) | 8 of 109 takes (7.3%)
- Walking False Alarms | 45.4 / hour (6 in 0.132 h) | 30.3 / hour (4 in 0.132 h)
- Still-Take False Alarms | 3.2 / hour (1 in 0.314 h) | 6.4 / hour (2 in 0.314 h)
- Median Trigger Latency | 2.05 s post-onset | 1.99 s post-onset

Detailed Take-Level Breakdown

Per-Fold Performance & Caveats

- Per-Fold Falls Caught: S1: 2/8, S2: 2/5, S3: 4/5, S4: 3/8, S5: 5/8, S6: 5/8 (Total: 21/42).
- Look-Alike False Alarms: 5 in sit_down, 3 in stand_up, 0 in bend_pick.
- Honest Engineering Reality: Catching 50% of falls with a 7.3% look-alike false alarm rate and 30 walking alarms/hour demonstrates proof-of-concept feasibility, but confirms the system is not yet a clinical medical device.

ABLATION ANALYSIS

Systematic Ablation Studies: What Drives Physical Sensing?

Speaker 3

Feature & Sensor Ablations (Events Task, SVM-RBF, 4 s)

Macro-F1 Performance Impact

- Ablation Condition | Macro-F1 | Delta vs Reference | Physical Meaning
- ---
- Dynamic Reference | 0.539 | Reference | All 50 dynamic f2 features
- Static Only | 0.353 | -0.186 | Static profiles fail completely
- RX1 Only (Upper) | 0.384 | -0.155 | Loses spatial diversity
- RX2 Only (Lower) | 0.403 | -0.136 | Loses link diversity
- Minus Motion Energy| 0.474 | -0.065 | Motion energy is top feature
- Minus Bandpowers | 0.494 | -0.045 | Doppler spectrum is critical
- Minus Phase | 0.527 | -0.012 | Sanitized phase adds boost
- Amplitude Only | 0.535 | -0.004 | Phase helps slightly on events

Key Physical Insights

Core Takeaways

- Dual Antennas are Mandatory: Dropping either receiver collapses F1 from 0.539 to ~0.39. Two spatial vantage points are essential to overcome human body shadowing.
- Doppler & Energy Dominate: Motion energy and FFT bandpowers carry over 70% of discriminatory event information.
- Phase Utility: Phase adds minor gains (+0.012) to events, but substantial stability (+0.039) to presence.

Live Real-Time Streaming Engine Architecture (app/engine_proc.py)

Speaker 4

Decoupled Multi-Process Architecture (D-057)

Overcoming Streamlit GIL Contention

• The Problem: In initial testing, running Streamlit web GUI and CSI serial ingestion in a single process caused GIL lock contention, degrading engine step rate from 0.50 s to 1.45 s (p90 2.27 s).
• The Fix (D-057): Separated into a decoupled two-process architecture:
• 1. Engine Daemon (app/engine_proc.py): Native background process reading serial streams, maintaining circular ring buffers, and executing inference every 500 ms.
• 2. GUI Dashboard (app/live_app.py): Streamlit web interface polling predictions via local authenticated IPC socket.
• Outcome: Live step compute locked at 383 ms median, stepping precisely every 0.50 s.

Safety Features & Zero Stale Window Protection (D-053)

Defensive Live Runtime

• Stale Buffer Detection: If any receiver falls silent for > 1.0 s, engine outputs 'degraded' or 'no_signal', NEVER predicting on stale packets.
• Auto-Reconnection: Recovers dynamically via /dev/serial/by-id hardware tokens on unplug/replug.
• BLAS Single-Threading: Limited OpenBLAS to 1 thread on host laptop, slashing step compute from 533 ms to 251 ms.

LIVE HARDWARE DEBUGGING

Live Library Field Trials & The RX2 Connector Diagnostic

Speaker 4

The Morning Live Mystery (Session 3)

Complete Loss of Motion Response

• The Symptom: Live test ran for 14 min without crashing, but walking across the link was called 'empty' (walk 3/83) and the fall produced zero alarm (max P = 0.33).
• Offline Verification: Diagnostic takes recorded on the boards failed identically offline, proving the software was innocent—the issue was physical RF!
• The Anomaly: RX2 RSSI was 4.5 dB abnormally high (-33.6 dBm vs -38 dBm baseline), yet RX2 motion features sat at the 0th to 5th percentile (completely motion-blind!).

The Antenna Swap Test & U.FL Resolution (D-060)

Systematic Hardware Isolation

• Antenna Swap (Round 3): Swapped Antenna A (from RX1) to RX2, and Antenna B to RX1.
• The Discovery: RX2 immediately sprang to life! Motion features surged to the 72nd–97th percentile. Walking was recognized 47/52 times. RX1 remained healthy with Antenna B.
• Root Cause Identified: A poorly mated U.FL micro-coaxial connector on RX2. It created RF reflection and severe static leakage that blinded the receiver to dynamic Doppler shifts.
• Resolution: Firmly reseated the U.FL connector and confirmed vertical polarization, restoring normal sensing.

LIVE SYSTEM VALIDATION

Final 14-Minute Continuous Live Field Validation Results

Speaker 4

Continuous Live Run Overview (2026-10-23 11:40–11:53)

1658 Steps at 0.50 s Interval

• Physical Setup: College Library Reading Hall, campus Wi-Fi active, spoken audio cues.
• System Health: Ran continuously for 14 minutes (1658 inference cycles) with zero crashes.
• Walk Across Link (Path 2, Horizontal): 37 of 42 windows correctly identified as Walk (88.1% recall)! Triggered 2 false fall alarms during fast walking (P = 0.57, 0.50).
• Walk Along Link (Path 1, Vertical): 6 of 51 windows correct (mostly called 'empty'; reveals Doppler blind zone).
• Simulated Mattress Fall: Event detected in 2 windows, but max P(fall) = 0.28 (failed to cross θ = 0.50 threshold -> NO ALARM).

Analytical Assessment of Live Reality

Why Live Diverged from LOSO

• 1. Presence Fragility: Logistic regression presence boundary sits inside empty-room variance; empty room was called occupied in 17/52 windows, corrupting state transitions.
• 2. Doppler Geometric Blind Spot: Moving along the line of sight (Path 1) produces minimal path-length changes, blinding the single-axis link.
• 3. RF Co-Channel Traffic: Campus library APs operating on Channel 11 caused burst packet loss up to 8%.

Systemic Limitations & Open Engineering Challenges

Speaker 4

Scope Limitations (Mandatory Compliance)

Boundary Conditions of Study

- 1. Confined Physical Domain: All experimental data gathered in a single room (library hall) with fixed furniture.
- 2. Demographic Homogeneity: 6 healthy university volunteers (young adults); gait kinematics and body mass do not reflect frail geriatric demographics.
- 3. Simulated Fall Constraints: Falls performed onto a high-density mattress at a single spot (Spot M); cannot prove fall detection elsewhere in the room.
- 4. Ground Truth Proxies: No multi-camera optical motion-capture ground truth; timing relies on audio countdowns and DSP energy onset proxies.
- 5. Non-Clinical Device: Prototype is an academic proof-of-concept, not a certified life-safety device.

Core Technical Deficiencies

Engineering Bottlenecks

- Presence Baseline Drift: Static profile shifts cause the presence classifier to misidentify empty rooms as occupied.
- Directional Blind Zones: Single-axis LoS links struggle to detect motion parallel to the radio beam.
- Hardware RF Vulnerability: U.FL connector mating can silently blind Doppler sensitivity without triggering an RSSI drop.

Academic Conclusion, Future Roadmap & Viva Voce Defense

Speaker 4

Key Project Achievements

Summary of Deliverables

• • Built End-to-End Testbed: Low-cost COTS ESP32-S3 transceivers capturing 114 subcarriers at 200 Hz.
• • Rigorous DSP Pipeline: Created anti-aliased, gain-normalized, phase-sanitized feature space f2.
• • Unbiased Evaluation: Proved classical models beat deep learning on small CSI data (F1: 0.70 presence, 0.57 state, 0.54 events; 21/42 falls caught).
• • 100% Reproducible: Automated script rebuilds features, retrains models, and validates 801 raw file checksums in 11 minutes (Phase 6 Certified).

Future Technical Roadmap

Next-Generation Enhancements

• 1. Guided Site Calibration: 60-second automated empty-room baseline capture on startup (docs/analysis/calibration_proposal.md).
• 2. Multi-Angle Spatial Diversity: Add a 3rd receiver at 90° azimuth to eliminate geometric Doppler blind spots.
• 3. Automated Hardware Health Check: Implement preflight loopback tests to detect unseated RF connectors.
• Thank you! We now welcome questions from the examination committee.